DepthWizard — Demo Video Script

SIH 2026 · PS **SIH26175** (ISRO / SAC) · Theme **Disaster Management**. Screen recording of the local prototype, narrated live. Target **7–8 minutes**.
Green = what you **do** · Bordered = what you **say** · Italic cues are transitions, say them so the video flows as one argument rather than a list of features.

> **THE ARGUMENT THE VIDEO MAKES, IN ORDER** **1.** Elevation is what you need after a disaster and what you never have. **2.** One photograph is enough — here it is working, live, on data it has never seen. **3.** Here is why it works, and why the obvious approach fails. **4.** Do not take my word for it: here is the airborne LiDAR beside it. **5.** Here is every number, including the ones that are not flattering. **6.** Here is what it cannot do, and why we did not paper over it.

Three locations, deliberately

Beat	Scene	Why
Open 0:00	**Washington DC — civic** bundled, GAMUS	The most striking thing we have. Sets the bar before a word about method.
Upload 0:45	**valkenburg_hills** you upload live	Real relief, not flat polder — absolute elevation and the slope layer both have something to show, and it is a GeoTIFF so it exercises the georeferenced path.
Proof 4:15	the job you just ran, then **Philadelphia**	Comparison on something unprepared, then a second country and sensor.
Close 7:00	**Veluwe — forest** bundled, AHN	Canopy is where the limitation lives, so the admission has the right picture behind it.

Before you hit record

Step	Why
`cd heimdall-web && npm run dev` in **your own terminal**	Nothing else can then stop the server mid-take.
GSD override box empty	It applies only to images with no coordinate system. A leftover value overrides a GeoTIFF's own scale and puts a warning banner on screen.
Pre-run `valkenburg_hills`, leave it in Recent jobs	Insurance. If the live run stalls, cut to it with no dead air.
Open DC, orbit once, reload	Warms WebGL so the first frame is smooth.
Full screen · 100% zoom · notifications off	An ntfy popup costs a re-record.

> **One rule throughout: never say a number you cannot show.** Every figure here is measured and lives in the repository. If a judge pauses and checks, it holds up — that is the impression worth more than any adjective.

0:00 – 0:45 · The problem

Do: Open with **Washington DC — civic** in the 3D flythrough. Orbit slowly once, all the way round, while you talk. Touch nothing else.

SAY

When a flood or an earthquake happens, the first question is about height. How deep does the water get. Which blocks are above the line. Where can relief actually get through.

That needs an elevation model, and an elevation model is exactly what you do not have. Airborne LiDAR is a commissioned flight — weeks and a budget, and out of date the moment the ground changes. What you almost always *do* have, within hours, is a single satellite photograph.

SAY

This is DepthWizard. One ordinary RGB image in — no stereo pair, no LiDAR — and the surface comes out in real metres. What you are looking at is a district of Washington DC, rebuilt from a single photograph.

Let me show you it working on something it has never seen.

Avoid "AI-powered", "cutting edge", "state of the art". The people watching build satellites. Show them a measurement and they trust you; use marketing language and they stop listening.

0:45 – 1:30 · Upload something real

Do: Click **New processing job**. Drag `valkenburg_hills__UPLOAD_THIS.tif` onto the drop zone. Pause on the thumbnail. Check the **GSD override is empty**. Press **Process image**.

SAY

A real aerial GeoTIFF of Valkenburg, a hill town in South Limburg. Four hundred metres across, twenty-five centimetres per pixel — one of eight sites we deliberately hold out of training.

I am uploading it the way an analyst would. No preprocessing, no parameters. The file carries its own coordinate system, so the pipeline reads the scale from the file.

Do: Hover over **Reference DEM** and **Ground control points** so both are visible. Leave them empty.

SAY

Both optional, and both named in the problem statement. A local terrain model — SRTM, Copernicus, CartoDEM — or surveyed control points; one control point is worth about half a metre of absolute accuracy.

I am leaving them empty to show the default: everything it needs, it fetches itself. *While it runs — why this works, because the obvious approach does not.*

1:30 – 3:00 · Why it works

Do: Point at each stage as it lights up — **Ingest & georeferencing**, **Ground sample distance**, **Height estimation**, **Absolute calibration**, **GeoTIFF / PNG export**, **Textured 3D mesh**. Let the log scroll.

SAY

Foundation models predict depth from a single photograph very well — but *relative* depth. This is nearer than that. They were trained on ordinary photographs, where absolute distance is unknowable, so they have no idea what a metre is.

The whole problem is turning that into real elevation. The problem statement calls it scale calibration, and it is the hard part.

SAY

We freeze the backbone entirely and train a small head on top whose only job is that conversion. It learns from sixteen thousand tiles of imagery paired with measured LiDAR height, and it predicts a scale and a shift at *every pixel* — a roof and the street beside it need different corrections.

SAY

The obvious alternative is a *metric* depth model that outputs metres directly, skipping all this. We tried it first. It fails, and the reason is instructive.

Metric models assume a camera with perspective — things shrink with distance. An orthomosaic has none of that; every pixel looks straight down. We measured the correlation between what a metric model predicted and real LiDAR height: **0.06**. Indistinguishable from noise. The relative model, before we trained anything on it, reached **0.41**.

SAY

[OPTIONAL — cut this first if running long]

That single measurement is why the architecture looks the way it does, and it happens to match what the problem statement itself suggests: anchor scale-agnostic depth features to a coarse DEM. We use Copernicus GLO-30, downloaded automatically, so absolute elevation comes out without the user supplying anything at all.

IF ASKED FOR SPECIFICS Frozen Depth Anything V2 Base, 97 million parameters. Trainable head: 1.7 million, decoding the full feature pyramid. Trained on 16,004 tiles — GAMUS plus Dutch AHN LiDAR at two ground resolutions. Loss is scale-invariant log, plus multi-scale gradient matching to keep roof edges sharp, plus an L1 anchor, plus a height-weighted term. Shipped head: 6.6 MB.

3:00 – 4:15 · The result

Do: The job finishes and the **3D flythrough** opens. Orbit once slowly. Then switch to **Fly** and move along the hillside at roof height, slowly.

SAY

That is the surface, from the single image I uploaded ten seconds ago. True metres — no vertical exaggeration anywhere.

And because Valkenburg is on a hill, you can see the terrain doing real work: the ground rises, the buildings sit on it, and the two are separated.

Do: Click the **Elevation** layer, then **Above ground**, then **Slope**. Pause a beat on each. Return to **Optical**.

SAY

Three views of the same reconstruction. Absolute elevation, metres above sea level. Height above ground, with the terrain removed. And slope, computed from the surface.

Structural height and slope assessment from an arbitrary aerial view is named in the problem statement — and it is not a bolt-on, it falls out of having a real surface.

Do: Click **Measure**. Draw a line across the slope, over a couple of buildings. Let the elevation profile draw underneath.

SAY

Distance along the ground, and the height profile underneath it — measured on the reconstruction, not on the photograph.

Do: Click the **2D maps** tab. Scroll through the rasters. Return to **3D flythrough**.

SAY

And it is not only a picture. Every run writes standard georeferenced products — absolute DSM, the terrain model under it, height above ground, hill-shaded relief. Ordinary GeoTIFFs that open in QGIS with their coordinate system intact.

The 3D view is the interface. The raster is the deliverable.

Do: Tick **Flood / water level**, drag it up several metres, hold, untick.

SAY

And since this sits under disaster management — the water tool cuts a plane through the scene at any height and shows what goes under. On a hill town it reads immediately: the valley floods, the upper streets do not.

But you should not believe any of it just because it looks convincing. So let me prove it.

4:15 – 5:30 · The proof

This is the minute that wins or loses the video. Anyone can render a plausible surface. Almost nobody hands over the real measurement and invites the comparison. Slow right down here — this beat can afford to breathe.

Do: Scroll to **Validate against reference** below the viewer. Upload `valkenburg_hills__truth_heights.tif`. Let the metrics table and error map appear.

SAY

This is what I would want to see if I were evaluating.

I hand it the airborne LiDAR for exactly this ground and it scores itself — raw surface, datum offset removed, height above ground — and draws a map of where the error is, rather than reducing everything to one number.

Do: A **Compare against the reference** toggle appears above the panel. Flip **Heimdall ↔ Reference** two or three times, slowly, holding one camera angle.

SAY

And here is the surface I reconstructed a minute ago, flipped against the truth. Same grid, same texture, same vertical origin — only the heights differ.

The hillside tracks. Open ground lands within about twenty centimetres. And none of this was prepared — that is the file I dragged in at the start.

Do: Open **Example scenes → Philadelphia — riverside**. Toggle **Heimdall ↔ LiDAR truth** once, then leave it on Heimdall.

SAY

And the same on a US city from GAMUS, the dataset the problem statement recommends — different country, different sensor. *Which brings me to the numbers.*

5:30 – 6:45 · The numbers

Do: Click the **Accuracy** tab. Let the table fill the frame. Point at the mean row, then the landscape rows, then the resolution rows. Do not rush the pointer.

SAY

Every number behind that comparison is in the application, not on a slide.

Eight held-out Dutch sites against airborne LiDAR. Height above ground: **5.32 metres RMSE**, MAE **3.04**, correlation **0.67**. The honest reference is predicting nothing at all — a flat world — and that is **8.37**.

SAY

This problem is scored on the absolute DSM, terrain plus structures, so: **6.11 metres**. And half a metre of that is not our model — it is the constant offset between a global terrain datum and the local one. One ground control point drops it to **5.63**. That is why the field exists.

SAY

The problem statement asks for stability across urban, sparse, hilly and forested landscapes, so we measured each rather than hiding behind an average. Forest **2.24**, farmland **1.56**, hilly town **3.21** — the site I just uploaded — and dense urban **8.40**, the hardest case.

The ground itself, which is most of any image, comes in at **0.82 metres**.

Do: Scroll to the resolution rows.

SAY

Evaluation uses Cartosat 2S at sixty centimetres, so here is the same model on the same sites with the imagery degraded to match: **5.35** at twenty-five centimetres, **5.51** at sixty, **5.72** at one metre. Flat across the range that matters.

6:45 – 7:30 · How we know

Do: Stay on the Accuracy tab, or return to the 3D view — whichever reads better. Keep the pointer still.

SAY

One more thing, because it is what separates a demo from a result.

We ran six controlled experiments, changing one thing each time — capacity, data volume, the loss function. Only the loss mattered. Extra parameters did nothing; doubling the data did nothing to the headline figure.

And we ran one configuration twice on different random seeds to find out how much of an improvement was luck. About a tenth of a metre — which told us two of our earlier results were noise, and we reported them as noise.

SAY

Every site you have seen numbers for was excluded from training by a three kilometre radius. Nothing here is scored on data the model has seen.

7:30 – 8:15 · Limits, then close

> **Do:** Open **Example scenes → Veluwe — forest**. Orbit gently and keep orbiting to the end.

SAY

Which brings me to what it cannot do, because you will see it if you look closely.

Tall structures come out short — roughly half their height on the tallest buildings, and canopy behaves the same way, which is why I am showing a forest while I say it.

That is not a bug we have not got to. A single nadir photograph contains no parallax, so when the model cannot tell a ten-metre building from a twenty-metre one, predicting something in between is mathematically correct. We could scale the tall predictions up so the render looked better — and the measured error would get worse. So we do not.

SAY

Ground, roads and low vegetation stay sub-metre. Buildings and canopy carry almost all the rest, and we publish exactly how much.

All of this runs on a laptop from one command, CPU-only if there is no GPU. The trained head is six and a half megabytes, and every figure I quoted regenerates from a script in the repository — including the ones that are not flattering.

One photograph in. A measured, explorable 3D surface out. Thank you.

Delivery notes

- **Pace.** 1375 spoken words. At a normal delivery that is about **9 minutes 10**; speaking slowly and leaving room for the 3D to breathe, about **10 minutes 35**. Time yourself once before recording — the cut list below takes it under 8 if you need it.
- **The three moments that carry the video** are the metric-vs-relative measurement at 2:30, the reference toggle at 4:45, and the limitation at 7:30. Slow down for all three. Volunteering a weakness reads as confidence, and evaluators notice who does it.
- **Say the italic transition lines.** They are what turns six features into one argument. Without them the video is a tour; with them it builds.
- **Mouse discipline.** Hover a control for a beat before clicking, so the viewer's eye arrives before the change does. No fast spins — they read as showing off and compress badly.
- **If the live job fails on camera**, do not restart it. Say "I have one already processed" and click it in **Recent jobs**. Recovering smoothly looks better than a clean take.
- **Do not show** the terminal, the code, or the file system. Prototype only.
- **Running long?** Cut in this order: the dashed OPTIONAL block at 2:30, then the 2D maps beat at 3:45, then the "How we know" section at 6:45. Never cut the toggle or the limitation.
- **Running short?** Let the flythrough breathe. Silence over a moving 3D scene is fine and looks confident; filler talk does not.

Questions you may get afterwards

Question	Answer
"How can one image give height at all?"	The scale is not in the image — it is learned. The backbone gives relative ordering; the head converts that to metres from patterns learned across 16,004 LiDAR-measured tiles: shadow, texture gradient, apparent size, context. Where the image is genuinely ambiguous we under-predict, and we publish that bias.
"Why not a model that outputs metres directly?"	We tried. Metric models assume a perspective camera; an orthomosaic has none. Correlation with true height 0.06, against 0.41 for relative depth before any training.
"5.32 metres sounds like a lot."	It is dominated by tall structures. Ground and low vegetation are sub-metre. Predicting nothing scores 8.37. And no single-view method resolves a tower's height from one nadir look — we quantify the gap rather than hide it.
"Have you tested on ISRO imagery?"	Not yet — we have no ISRO LiDAR to score against, and the FAQ says reference data will not be provided. What we have tested is resolution: 5.51 m at the 0.6 m Cartosat 2S resolution evaluation will use.
"What happens above 2 m resolution?"	It degrades, and we say so. By 10 m a building spans one to three pixels and there is nothing left to infer from. The pipeline warns when the input is outside the range it was trained for.
"Does it need a GPU?"	No. CPU fallback is built in; `run.sh` detects the hardware and installs the matching build.
"Is this reproducible?"	Every number comes from a JSON file in the repository, and a script verifies the documentation against those files and fails if anything has drifted.
"What is the licence position?"	The default backbone is CC-BY-NC — fine for evaluation, not for commercial deployment. A permissively licensed alternative is measured as statistically equivalent and documented.

Every figure, with its source

Figure	Value
Height above ground, 8 held-out LiDAR sites	**5.32 m** RMSE · 3.04 MAE · r 0.67
Absolute DSM — what the setters say they grade	**6.11 m**, or 5.63 with one control point
Predicting zero height everywhere	8.37 m
Ground · low vegetation	0.82 m · 0.86 m
Forest · farmland · hilly · dense urban	2.24 · 1.56 · 3.21 · 8.40 m
0.25 m · **0.6 m (Cartosat 2S)** · 1 m	5.35 · **5.51** · 5.72 m
GAMUS test split, 703 tiles	4.74 m, against 8.59 m predicting nothing
Metric vs relative backbone, correlation with true height	0.06 vs 0.41
Seed-to-seed variation between identical runs	≈0.1 m
Valkenburg, the site uploaded live	3.21 m RMSE · r 0.73 · 5.38 m predicting nothing
Model · training set · inference	1.7 M trainable on frozen 97 M · 16,004 tiles · 6.6 MB · ~4.5 s per 400 m site

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Regenerate with `python scripts/benchmark_ahn.py`. Figures read from `outputs/benchmark_ahn/results.json`, `outputs/evaluation_envelope.json` and `outputs/gamus_test.json`. Repository: github.com/Aryan241/Heimdall